

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I : (Maximum Marks: 32)

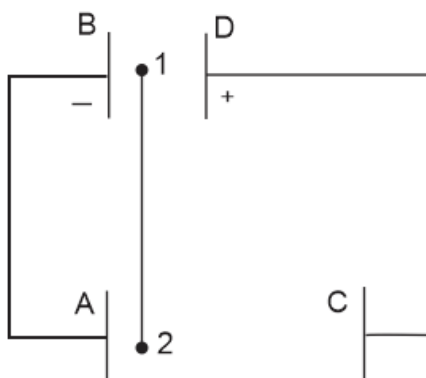
- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

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Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

1. In the situation shown the capacitors are charged and plates A and B are on the same vertical plane but plates C and D are on different vertical plane. Points 1 and 2 are on the same vertical plane inside the capacitor. If V_1 and V_2 be the electric potential and P_1 and P_2 be the gravitational potential at points 1 and 2 respectively then



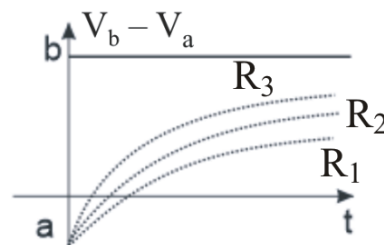
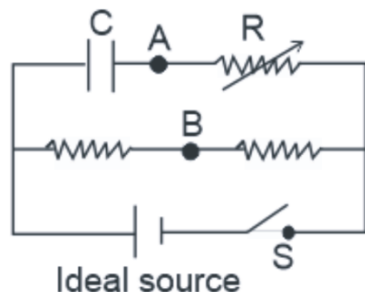
(A) $V_1 > V_2$ and $P_2 > P_1$

(B) $V_1 < V_2$ and $P_1 > P_2$

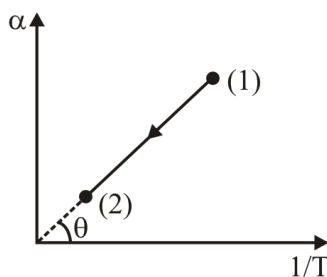
(C) $V_1 > V_2$ and $P_1 > P_2$

(D) $V_1 = V_2$ and $P_1 > P_2$

2. An uncharged capacitor C and a variable resistance R are connected to an ideal source and two resistors with the help of a switch at $t = 0$ as shown in the figure. Initially capacitor is uncharged and switch s is closed at $t = 0$ sec. The graph between $V_b - V_a$ for variable resistance R for its three different values R_1 , R_2 and R_3 versus time is shown. Choose the correct statement.



- (A) $R_1 < R_2 < R_3$ (B) $R_1 > R_2 > R_3$ (C) $|a| < |b|$ (D) $|a| > |b|$
3. According to the given graph match the correct statements, given below line gives the thermodynamics process and α is volume coefficient and T is the temperature of the ideal gas



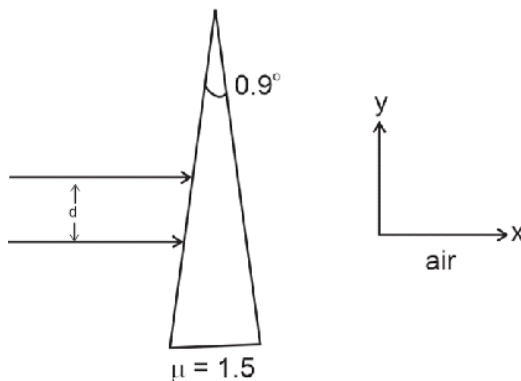
- (A) If $\theta = 45^\circ$, then the given process is isobaric process
- (B) If $\theta = 37^\circ$, then the pressure, volume and temperature of the gas are increasing from point (1) to point (2)
- (C) If $\theta = 53^\circ$, then the equation of the process is $T^4 V^{-3} = \text{constant}$
- (D) If $\theta = 53^\circ$ and the horizontal axis of the graph is taking T instead of $\frac{1}{T}$ then the equation of the process is $T^4 V^{-3} = \text{constant}$.

Space for Rough Work

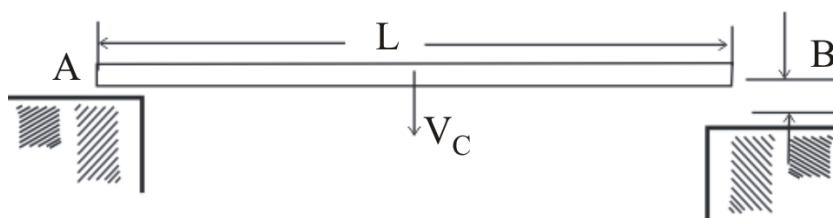
4. A radioactive sample has a half life of 40 seconds. When its activity is measured 80 seconds after the beginning, it is found to be 6.932×10^{18} dps. During this time from beginning to 80 sec, total energy released is 6×10^8 joule ($\ln 2 = 0.6932$) :-
- (A) The initial number of atoms in the sample is 1.6×10^{20}
- (B) The initial number of atoms in the sample is 1.6×10^{21}
- (C) Energy released per fission is 5×10^{-13} J
- (D) Energy released per fission is $\frac{5}{3} \times 10^{-13}$ J
5. Two identical discharge tubes are operated from the mains (230 V and 50 Hz) and they are both connected to inductors in series. The inductors are identical. A condenser is connected in series to one of the discharge tubes. With this arrangement, it is ensured that when the current through one of the tubes is maximum, the current through the other is minimum and vice-versa. In both branch, the average value of power is 8W. Which of the following statement(s) are correct ?
- (A) The value of resistance of discharge tube is nearly 3.3 k Ω
- (B) The capacitance of condenser is 6.6 μ F
- (C) The total impedance of the discharge tube and the inductor together is nearly 4.67 k Ω
- (D) The capacitive reactance is double of inductive reactance
6. Solar constant α is defined as intensity of solar radiation incident on earth. Its value is close to 1.4 kW/m². Nearly 75% of this energy is absorbed by the Earth. The average temperature of the Earth is about 300 K. Radius of the Earth is $R_E = 6400$ km and that of the Sun is $R_S = 700$ Mm. Earth-Sun distance is $r = 1.5 \times 10^8$ km. Assume Sun to be a black body, then mark the correct statements $\sigma = (5.67 \times 10^{-8} \text{ W/m}^2 \text{ K}^4)$, (Given : $(0.579)^4 = 0.113$)
- (A) The effective emissivity of Earth is nearly 0.6
- (B) The surface temperature of the Sun is nearly 5800 K
- (C) The effective emissivity of the Earth is 0.3
- (D) Amount of heat radiated by the Earth is nearly 135×10^{15} W

Space for Rough Work

7. Consider a beam of light of width d wavelength $\lambda = 221 \text{ nm}$ and power 7.2 W incident on a prism of angle 0.9° . Refractive index of prism is $\mu = 1.5$ and it is in air then which of the following option(s) is/are correct :



- (A) The angle of deviation is $\frac{\pi}{400} \text{ rad}$
- (B) The number of photons incident on the prism/time is $8 \times 10^{18} / \text{sec}$ approximately
- (C) The magnitude of force on prism by light beam is approximately $6\pi \times 10^{-11} \text{ N}$. (Neglect partial reflection)
- (D) Width of beam decreases after it passes through prism.
8. A uniform slender rod of length L is dropped onto rigid supports at A and B. Immediately before striking A the velocity of the rod is V_C . Since support B is slightly lower than support A, the rod strikes A before it strikes B. Assume perfectly elastic impact at both A and B. Let the angular velocities of the rod be ω_1 and ω_2 and the velocities of its C.M. be v_1 and v_2 immediately after the rod strikes at A and B respectively. Then (Consider only two collisions one at A and other at B)



- (A) ω_1 is clockwise
- (B) ω_2 is anticlockwise
- (C) v_1 is downward
- (D) v_2 is upward

Space for Rough Work

SECTION-II (i) : (Maximum Marks: 12)

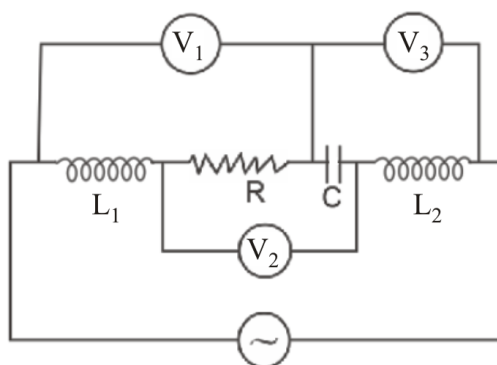
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- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
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Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

Reading of ideal hot wire voltmeter V_1 , V_2 and V_3 are 15, 20, 9 volt respectively. Source emf is 12 V. Z is net impedance of circuit and X_C is capacitive reactance.



1. Find $\frac{L_1}{L_2}$.
2. Find $\frac{X_C}{R}$

Space for Rough Work

Paragraph for Questions 3 and 4

The YDSE apparatus is modified by placing an isotropic transparent plate of high melting point in front of one of the slits. The refractive index of the plate is $\mu = 1.5$ at room temperature and its thickness is $t = 2 \mu\text{m}$. The refractive index of plate will increase when temperature increases and temperature coefficient of refractive index of plate (i.e. the fractional change in refractive index per unit rise in temperature) is $3 \times 10^{-3}/^{\circ}\text{C}$. The incident light is having wavelength $\lambda = 6000 \text{ \AA}$. The separation between the slits is $d = 0.2 \text{ cm}$, and separation between the slits and the screen is 2m . Based on above information, answer the following questions.

3. If the plate is heated so that its temperature rises by 100°C , then how many more fringes will cross a particular point on the screen as compared between before & after heating? (Neglect the thermal expansion of plate)
4. If intensity of uncovered slit is I_0 and that of covered slit is $\frac{I_0}{2}$, then intensity at midpoint after heating of the screen is CI_0 then value of C is

Paragraph for Questions 5 and 6

The separation between the objective and the eye piece of a compound microscope can be adjusted between 9.8 cm to 11.8 cm . Focal length of the objective and the eyepiece are 1.0 cm and 6cm respectively. Eyepiece is movable and image is always needed at 24 cm from the eye. $D = 24 \text{ cm}$.

5. Find maximum magnifying power which can be produced by the microscope.
6. Separation between the lenses for minimum magnifying power is (in cm)

Space for Rough Work

SECTION-II (ii) : (Maximum Marks: 16)

- This section contains **FOUR (04)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
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-
7. A student perform an experiment to determine the Young's modulus of a wire, exactly 2m long by Searle's method. In a particular reading, the student measures the extension in the length of the wire to be 0.8mm with an uncertainty of ± 0.05 mm at a load of exactly 1.0kg. The student also measures the diameter of the wire to be 0.4mm with an uncertainty of ± 0.01 mm. Take $g = 9.8\text{m/s}^2$ (exact). Find the maximum percentage error in the measurement of young's modulus of the material of the wire from these data.
 8. Electric field exists in a region given by $4(1 - 2x)\hat{i}$ N/C. Find the minimum velocity V_0 (in m/s) with which a particle of mass m and charge- q should be projected from origin in the positive x -direction so that it reaches to infinity. Specific charge of particle is 8.
 9. A closed organ pipe of length 99.4 cm is vibrating in its first overtone and is always in resonance with a tuning fork having frequency $f = 2(150 - t)$ Hz, where t is time in second. The rate by which radius of organ pipe changes when its radius is 1 cm, is $1/(9x)\text{ms}^{-1}$. Find x (speed of sound in organ pipe = 320 m/s)
 10. A nuclear reactor containing U^{235} produces 10 MW. If the fission rate assuming approx 200 MeV of useful energy is released in one fission is $K \times 10^{17}\text{sec}^{-1}$ find the value of K .
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Space for Rough Work

PART-2 : CHEMISTRY
SECTION-I : (Maximum Marks: 32)

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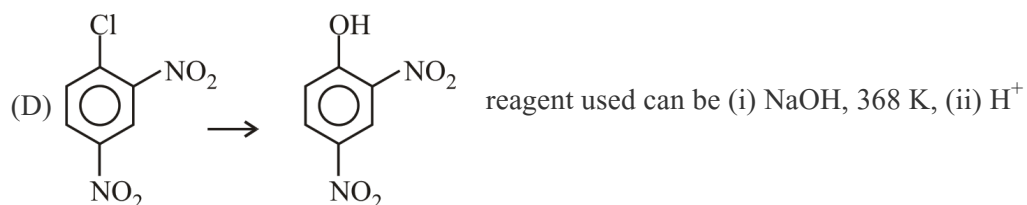
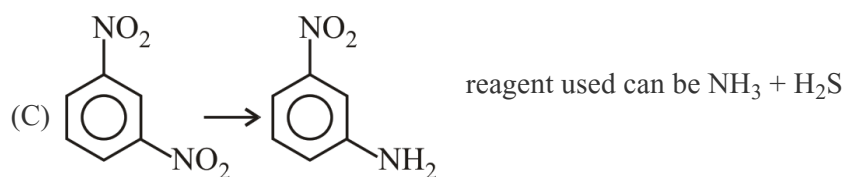
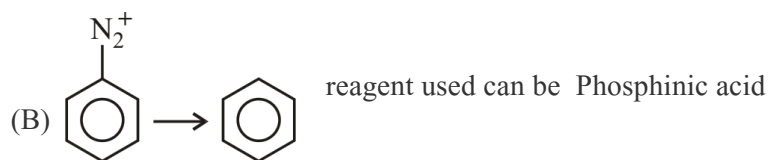
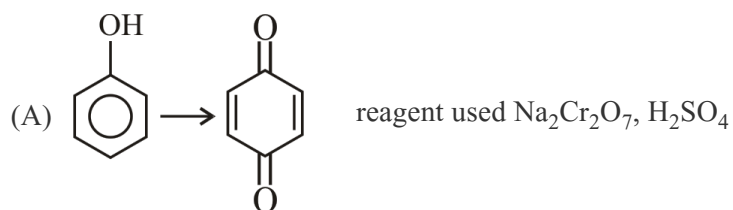
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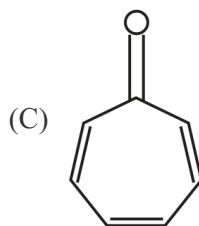
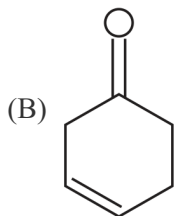
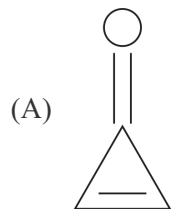
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1. 0.5 moles of sodium chloride is dissolved in 150 moles of water. The solution is cooled to a temperature 'T' K below the freezing point of solution so that some ice is formed. After separating the ice formed, the remaining solution is heated to normal boiling point of water, where the vapour pressure of solution is found to be 750 torr. Identify correct option(s) on basis of this information : (K_f for water = 2 K kg mol^{-1})
- (A) Mass of ice formed at 'T' K is 1.350 kg.
- (B) Temperature T is $\left(273 - \frac{40}{27}\right) \text{ K}$
- (C) Freezing point of the original solution is $\left(273 - \frac{20}{27}\right) \text{ K}$
- (D) Relative lowering of vapour pressure of resulting solution is $\frac{1}{76}$
-

Space for Rough Work

2. For the following conversion the reagent used can be



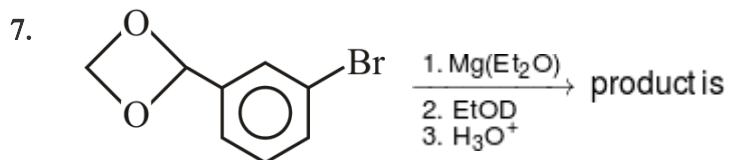
3. Which of the following will have $\text{C}=\text{O}$ bond length almost similar to $\text{C}-\text{O}$ single bond length ?



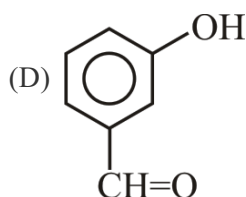
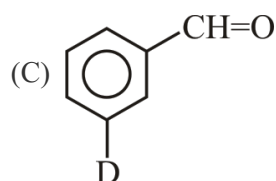
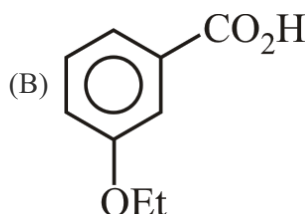
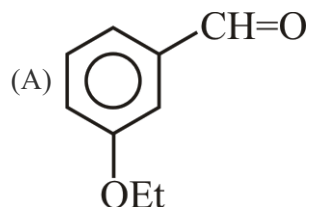
Space for Rough Work

4. Which of the following are incorrectly matched ?
- (A) Cassiterite - SnO_2
 - (B) Malachite - $\text{CuCO}_3\text{Cu(OH)}_2$
 - (C) Azurite - Cu_2O
 - (D) Copper glance - Cu_2S
5. Which of the following statements is/are incorrect about XeF_6 ?
- (A) It can be stored in glass vessel.
 - (B) On hydrolysis with water it forms only XeOF_4 .
 - (C) XeF_6 acts as fluoride acceptor in HF to form HXeF_7 .
 - (D) With BF_3 it forms an addition product $[\text{XeF}_5^+][\text{BF}_4^-]$
6. Which of the following are biodegradable polymers ?
- | | |
|---------------------|------------|
| (A) Nylon 2-nylon 6 | (B) PHBV |
| (C) Dextron | (D) Melmac |
-

Space for Rough Work



Select the correct options :



8. An ester A ($C_4H_8O_2$) on treatment with excess CH_3MgBr followed by hydrolysis gives an alcohol B as the only organic product. Which of the following statements is/are true ?

- (A) A on basic hydrolysis gives formate ion as one of the product
 (B) B gives positive iodoform test
 (C) A on acid hydrolysis gives acetic acid as one of the products
 (D) B gives immediate precipitate with Lucas reagent

Space for Rough Work

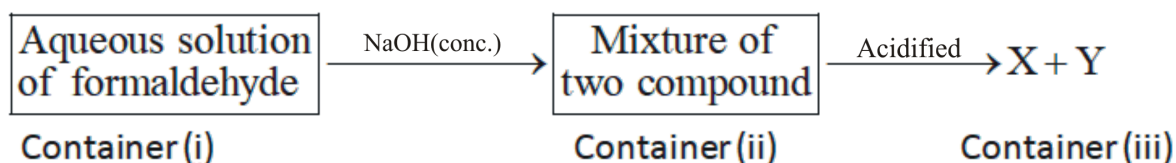
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Paragraph for Questions 1 and 2



pK_a of Y is 4.2 which is very small as compared to compound X. (Use all the data at 25°C).

Assume all above given reactions are occurring to completion.

1. If 0.2 M 100 ml aqueous solution of 'Y' mixed with 0.1 M 100 ml aqueous solution of formic acid and 0.1 M 200 ml aqueous solution of NaOH forming final aqueous solution (X). Find pH of (X) ?
2. If 0.1 M 50 ml NaCl and 0.1 M 50 ml Ca(OH)_2 mixed to the solution (X) formed above. Then what will be final pH of the solution ? [assume $(\text{HCOO})_2\text{Ca}$ is completely soluble in water]
[Given : $\log 2 = 0.3$, $\log 3 = 0.48$]

Space for Rough Work

Paragraph for Questions 3 and 4

In 1886, the French chemist, Francois Raoult after a series of experiments on a number of solvents including water, benzene and ether, succeeded in establishing a relationship between the lowering of vapour pressure of a solution and the mole fraction of the non volatile solute.

$$P_1 = X_1 P_1^0$$

Where P_1 is vapour pressure of solvent in an ideal solution and P_1^0 is vapour pressure of a pure solvent
 X_1 mole fraction of solvent and X_2 mole fraction of solute

$$X_1 + X_2 = 1$$

$$P_1 = (1 - X_2) P_1^0$$

$$\frac{P_1^0 - P_1}{P_1^0} = X_2$$

The relative lowering of vapour pressure of a solution containing a non-volatile solute is equal to the mole fraction of solute present in the solution.

3. The vapour pressure of pure liquid X and pure liquid Y are 32 and 102 mm of Hg respectively at 400 K. A liquid mixture of X and Y is composed of 1 mole each. What will be the pressure, when 1 mole of the mixture has been vapourized.
4. A solution of non-volatile solute in a mixture of solvent A and Solvent B has a vapour pressure of 40 mm of Hg at 20°C and 68 mm of Hg at 30°C. The vapour pressure of pure solvent (A) is 15 mm of Hg at 20°C and 34 mm of Hg at 30°C and the vapour pressure of pure solvent (B) is 85 mm of Hg at 20°C and 136 mm Hg at 30°C. Find the mole fraction of solvent A in the solution.

Space for Rough Work

Paragraph for Questions 5 and 6

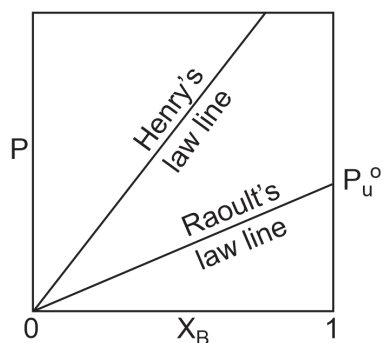
Raoult's law for solvent and Henry's law for solute

In ideal solutions, the solute as well as solvent obey Raoult's law. However, Henry found that, for real solutions at low concentrations, although the vapour pressure of the solute is proportional to its mole fraction, the constant of proportionality is not the vapour pressure of the pure substance but an empirical constant called Henry's constant. In a real solution Raoult's law is applicable to component which is in excess and Henry's law is applicable to component which is in very small amounts.

$$P_B = X_B K_B$$

X_B = mole fraction of solute

K_B = an empirical constant which is called as Henry's constant.



5. Henry's constants at 35°C-Chloroform : 175 Torr, Acetone : 150 Torr.
Vapour pressures of pure substances at 35°C-Chloroform : 270 Torr, Acetone : 347 Torr.
Calculate the vapour pressure of a solution of acetone in chloroform (in Torr) when the mole fraction of acetone is 0.05 ?
6. The total vapour pressure of a 4 mol% aqueous solution of Methanol in water at 293 K is 50 Torr and the vapour pressure of pure water is 20 Torr at this temperature. Calculate the total vapour pressure for a 5 mol% aqueous solution of Methanol in torr :

Space for Rough Work

SECTION-II (ii) : (Maximum Marks: 16)

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- Answer to each question will be evaluated according to the following marking scheme:

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7. How many of the following reagents can be used to detect $C_6H_5NH_2$
 $CHCl_3/KOH$, $CS_2/HgCl_2$, $C_6H_5SO_2Cl$, H_2O_2/OH^- , $K_2S_2O_8$, $FeCl_3$, $C_6H_5N_2^+Cl^-$
8. How many of these reaction gives N_2 as product
- (i) $NH_4NO_3 \xrightarrow{\Delta}$
- (ii) $NO \xrightarrow{\text{red hot copper}}$
- (iii) $Ba(N_3)_2 \xrightarrow{\text{heating}}$
- (iv) $NH_4Cl + NaNO_2 \xrightarrow{\Delta}$
- (v) $NH_2CONH_2 + HNO_2 \rightarrow$
- (vi) $NaNO_3 \xrightarrow{\text{heating, } 800^\circ C}$
- (vii) $NH_3 + Cl_{2(\text{less})} \rightarrow$
- (viii) $(NH_4)_2Cr_2O_7 \xrightarrow{\text{heating}}$
- (ix) $NH_4Cl + CaOCl_2 \rightarrow$
9. 3 mol of a mixture containing same number of moles of $NaNO_3$, NH_4NO_3 , $LiNO_3$, $Ca(NO_3)_2$, $Pb(NO_3)_2$ and $AgNO_3$, was heated strongly. Total number of moles of $NO_2(g)$ evolved is;
10. Molecular formula of an open chain compound(X) is C_8H_{14} . It decolourises bromine water. On reductive ozonolysis it forms acetaldehyde and butan-1, 4-dial in molar ratio 2 : 1. How many allylic carbon atoms are present in one molecule of compound (X) ?

Space for Rough Work

PART-3 : MATHEMATICS
SECTION-I : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
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Negative Marks : -2 In all other cases.

1. Let x, y, z, w are four positive real numbers satisfying the following system : $x + y + z + w = 12$ and $xyzw = 27 + xy + xz + xw + yz + yw + zw$, then which of the following is/are true ?

(A) $\frac{x+y}{y+z} = 1$

(B) $x - y + z \geq 2$

(C) $\frac{x}{y} - \frac{y}{z} > 2$

(D) $\frac{x+y-2z}{x+z} = 1$

2. If lines $x = a + m;$
 $y = -2;$
 $y = mx$ are concurrent (where m is negative), then the possible value(s) of 'a' is/are -

(A) 3

(B) $2\sqrt{2}$

(C) 2

(D) 0

Space for Rough Work

3. Which of the following statement(s) is/are correct ?

(A) $\int_{-1}^4 f(x) \cdot e^{-x^2+x+1} dx < 0 \Rightarrow$ Equation $f(x) = 0$ must have atleast one root in $(-1, 4)$

(B) $\int_0^1 \frac{\sin x}{x} dx < \int_0^1 \frac{\tan x}{x} dx$

(C) $\left| \int_a^b (f(x) + g(x)) dx \right| \leq \int_a^b (|f(x) + g(x)|) dx$

(D) $\int_0^{\infty} [2e^{-x}] dx = \int_0^{\ln 2} [2e^{-x}] dx = \int_0^{\ln 2} [e^x] dx$ (where $[.]$ denotes greatest integer function)

4. A fair die is rolled 'n' times. If $P(n)$ denotes the probability that there are atleast two equal numbers among the result obtained then-

(A) $P(n \geq 7) = 1$

(B) $P(n = 2) = \frac{1}{6}$

(C) $P(n = 4) = \frac{13}{18}$

(D) $P(n = 6) = \frac{317}{324}$

5. Given $A = \begin{bmatrix} 1 & \omega^3 & \omega^2 \\ \omega^3 & 1 & \omega \\ \omega^2 & \omega & 1 \end{bmatrix}$, then (where ω is non real cube root of unity)-

(A) A is a non-singular matrix

(B) A is an orthogonal matrix

(C) A^{-1} is a symmetric matrix

(D) $A^{-1} = \frac{1}{3} \begin{bmatrix} 1-\omega^2 & 0 & \omega-\omega^2 \\ 0 & 1-\omega & \omega^2-\omega \\ \omega-\omega^2 & \omega^2-\omega & 0 \end{bmatrix}$

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6. If $0 \leq \theta \leq \pi$, $i=1,2,3,4,5$ then roots of the equation $\sin^2\theta_1 z^4 + \cos^2\theta_2 z^3 + \sin^2\theta_3 z^2 + \cos^2\theta_4 z + \sin^2\theta_5 = 3$, satisfying $|z| < 1$, will lie
- (A) outside circle with radius $\frac{2}{3}$ and centre at origin
- (B) inside circle with radius 1 and centre at origin
- (C) outside circle with radius 1 and centre at origin
- (D) inside circle with radius $\frac{1}{2}$ and centre at origin
7. Let $N = ab$ be a two digit number (where $b \neq 0$) which is divisible by both a and b , then
- (A) number of such numbers N is 14
- (B) number of such numbers N is 18
- (C) maximum value of $\frac{b}{a}$ is 5
- (D) $a + b$ is always an even number
8. $f(x) = \frac{[x] + 1}{\{x\} + 1}$ for $f : [0, \frac{5}{2}) \rightarrow (\frac{1}{2}, 3]$ where $[.]$ represents greatest integer function and $\{.\}$ represents fractional part of x , then which of the following is true.
- (A) $f(x)$ is injective discontinuous function
- (B) $f(x)$ is surjective non differentiable function
- (C) $\min \left(\lim_{x \rightarrow 1^-} f(x), \lim_{x \rightarrow 1^+} f(x) \right) = f(1)$
- (D) $\max (x \text{ values of point of discontinuity}) = f(1)$

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SECTION-II (i) : (Maximum Marks: 12)

- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

Consider the following function $f(x) = \begin{cases} \sqrt{x} & x \geq 1 \\ x^3 & 0 \leq x < 1 \\ \frac{x^3}{3} - 4x & x < 0 \end{cases}$

1. Number of the solution of $f(x) + 2 = 0$
2. Number of points of discontinuity of $f(x)$?

Paragraph for Questions 3 and 4

Let $f(x) = x^2 + 2$ and $g(x) = \sqrt{x - 2}$

3. The minimum distance between the curves $y = f(x)$ and $y = g(x)$ is :
4. The minimum distance between the curve $y = f(x)$ and the line $y = 2x - 3$ is :

Paragraph for Questions 5 and 6

Let $A(1, 2, 3)$, $B(2, 1, 4)$, $C(a, a^2, 6)$ are coplanar points. L is a line passing through points A and B.

5. Sum of all possible values of 'a' is -
6. If $a \in \mathbb{I}$, then perpendicular distance of point C from line L is -

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SECTION-II (ii) : (Maximum Marks: 16)

- This section contains **FOUR (04)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

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7. An ant moves on the curves $S : x^2 + y^2 = 9$ and another ant moves on the curve $(x + 10)^2 + y^2 = 16$, then the minimum distance between the two ants.
8. Let 'e' be the eccentricity of a hyperbola and $f(e)$ be the eccentricity of its conjugate hyperbola. If $\int_2^6 f(f(\dots f(e))) \, de = k$ (n is odd), if the value of $76k^2 - k^4$ is 2^p , then p is
9. A parabola touches the x and the y-axis at (1, 0) and (0, 1). If L denotes length of latus rectum of the parabola, then L^2 is.
10. $y = f(x)$ is a solution of differential equation $\frac{dy}{dx} + y = \frac{2xe^{-x}}{1 + ye^x}$ such that $f(0) = 1$, then $\left[\frac{f(-1)}{e} \right]$ is equal to {where $[.]$ greatest integer function}
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NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them._____
Signature of the CandidateI have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same._____
Signature of the Invigilator

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